MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG2DSCMOS100 **- Foundation of Mobile Development Systems**

Duration: 2 hour **Maximum Marks: 35**

PracticalExamination–ModelQuestionPaper

Instructions:

- **Demonstrate**thegiventask
- **Viva session** (Minimum 5 questions from the PracticalModule)
- **LabReport**

EvaluationCriteria:

- **Viva**:7 marks
- **LabReport**:8 marks
- **Demonstration**:20marks

Tasks:

Dart programs*(Chooseanyone)*

1. A dart program to find a given number is odd or even

2. A dart program to find the given number is Armstrong or not

3. A dart program to find the sum of N numbers

4. A dart program to find the prime number series less than N

5. Write a dart program which will do the following tasks

   1.Create an empty list

   2. Add items to empty list

   3.Add an item in a specific location

   4.remove from list

   5.remove from specific location

6.Define a class Student with properties name, age, and grade. Create an object of the class and display the student's details.

7.Implement a Car class with attributes brand, model, and year. Add a method displayCarInfo() to show the car's details.

8.Create a Map to store student names as keys and their marks as values. Write a program to display all students who scored more than 75 marks.

9.Create a Map that stores country names as keys and their capital cities as value

10.Implement single inheritance: Create a Parent class with a method sayHello() and derive a Child class that calls this method.

Flutter programs*(Chooseanyone)*

1. A flutter program to display simple message using stateless widgets
2. A flutter program to Generate increment counter by pressing a button using statefull widgets
3. A flutter program to display an image using assets
4. Login Page with Form Validation & Styling

    Task: Create a Flutter login page with a form containing email and password fields.

    a.Implement validation (e.g., email format, password length).

    b.Style the text fields with curved borders.

    c.Add a styled login button.

5. List of Fruits with Detail Screen

    Task: Create a Flutter app that displays a list of fruits, each with an image and name.

    a.Clicking on a fruit should navigate to a new screen displaying the same fruit's details.

    b.Use a ListView.builder() to display the list dynamically.

6. Firebase Authentication (Basic)

    Task: Create a Flutter app that integrates Firebase Authentication.

    a.Allow users to sign up and log in with email and password.

    b.Display a welcome message after a successful login.

7. Fetch Data from API & Handle Errors

    Task: Use the https://jsonplaceholder.typicode.com/ API to fetch and display data in a Flutter app.

    a.Show a list of users/posts in a ListView.

    b.Implement error handling for failed API calls (e.g., show an error message).

8. UI Clone Task

    Task: Choose a simple UI design and replicate it in Flutter.

    It can be a login page, profile page, or any basic screen with proper layout and styling.